

Lem Invariancia Conforme Longitud Hiperbolica

Lema 1. Sea Γ un camino en $\mathbb{D}$ y $f \in \text{Aut}(\mathbb{D})$, entonces $\ell(f \circ \Gamma) = \ell(\Gamma)$.

Demostración: Como $f \in \text{Aut}(\mathbb{D})$, $f \circ \Gamma$ es un camino en $\mathbb{D}$.

Por el Teo-formula-aut-disco-unidad/??, $\exists \gamma \in \mathbb{T}, w \in \mathbb{D} : f = \rho_\gamma \circ \tau_w$. Por tanto, para el integrando de la longitud hiperbólica de $f \circ \Gamma$ se tiene que

$$\frac{|(f \circ \Gamma)'(t)|}{1 - |(f \circ \Gamma)(t)|^2} = \frac{|\gamma| |(\tau_w \circ \Gamma)'(t)|}{1 - |\gamma|^2 |(\tau_w \circ \Gamma)(t)|^2} = \frac{|(\tau_w \circ \Gamma)'(t)|}{1 - |(\tau_w \circ \Gamma)(t)|^2} = \frac{|\tau_w'(\Gamma(t))| |\Gamma'(t)|}{1 - |\tau_w(\Gamma(t))|^2}.$$

Ahora, por el teorema de Schwarz-Pick, sabemos que $|\tau_w'(z)| = \frac{1 - |\tau_w(z)|^2}{1 - |z|^2}$, luego

$$\frac{|(f \circ \Gamma)'(t)|}{1 - |(f \circ \Gamma)(t)|^2} = \frac{|\Gamma'(t)|}{1 - |\Gamma(t)|^2} \implies \ell(f \circ \Gamma) = \ell(\Gamma). \quad \blacksquare$$

Referenciado en

- Teo Camino Minimo Poincare

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